Innovation, Competition, and Market Structure

A Cournot General Equilibrium Growth Model

PhD Reading Group | October 2026

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Motivation: the Schumpeterian trade-off

Creative destruction: market power is the reward for innovation.

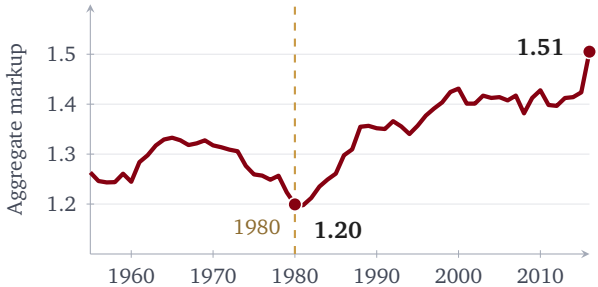
Monopoly rents fund R&D, and the innovations they pay for sustain growth (Aghion and Howitt, 1992; Grossman and Helpman, 1991).

Static efficiency. More competition, lower markups: prices closer to marginal cost.

Dynamic efficiency. Some market power gives firms a reason to innovate: faster growth.

An economy may accept higher prices *today* for faster progress *tomorrow* (Aghion et al., 2001, 2005).

The puzzle: markups up, dynamism down



Aggregate markup, weighted by total cost: U.S. listed firms (De Loecker et al., 2020).

- **Markups** up since 1980, whatever the weights or sample (Benkard et al., 2026).
- **Concentration** up, **labour share** down (Autor et al., 2020).
- **Business dynamism** down (Akcigit and Ates, 2023).

The question. If rents are what pay for innovation, why has rising market power coincided with weaker dynamism and slower growth?

The literature, and what it leaves out

Recent answers: ideas are harder to find, diffusion is slower, interest rates are low, intangibles and superstar firms (Bloom et al., 2020; Akcigit and Ates, 2023; Liu et al., 2022; De Ridder, 2024; Aghion et al., 2023).

To stay tractable, they simplify competition

- One producer per product line, with limit pricing (Klette and Kortum, 2004; Peters, 2020).
- A duopoly, or a fixed set of rivals (Aghion et al., 2001; Hopenhayn and Okumura, 2026).
- Oligopolists plus a fringe, with parametric entry and exit (Cavenaile et al., 2026).

Missing 1. Entry and exit are imposed rates, not derived from strategic interaction.

Missing 2. *Who* produces never matters for growth: misallocation only lowers the level of output.

This paper: two modelling contributions

1 Entry and exit as corner solutions

$$y_i > 0 \iff p > \frac{w}{A_i}$$

No fixed costs or entry rules: a firm produces if and only if the price covers its marginal cost. Entry and exit are equilibrium outcomes.

Fills Missing 1

2 Innovation relative to the technology in use

$$\eta_i = \eta\left(l_i^r,\ \frac{A_i}{\bar{A}}\right), \qquad \bar{A} = \frac{Y}{L^p}$$

The entrant is the least productive producer: entry lowers markups, but also $\bar{A}$, and so every firm's research. A dynamic cost of competition.

Fills Missing 2

The application: what changed in the U.S.?

1 Two regimes

Past (1953–1983) and Now (1999–2024), dated from the aggregate markup.

2 Two calibrations

The same model, fitted to each regime's markups, R&D and growth.

3 One comparison

Which parameters changed, and which change matters most?

Answer. The innovation technology. First order: θ, the elasticity of innovation to research, rose from 0.05 to 0.13. Second order: ε, the pull of catching up, fell from 2.0 to 1.2. Firms research more, get less per unit, and leaders stay ahead.

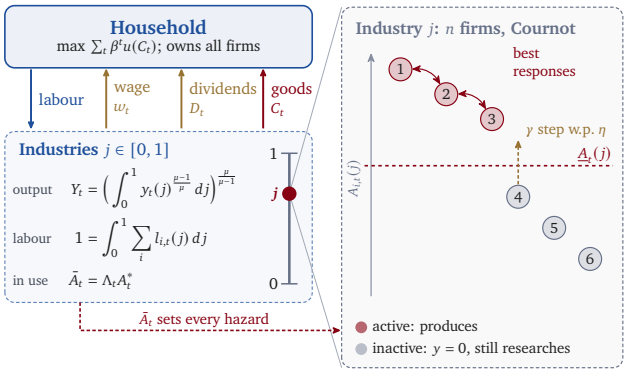
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The model in one picture



Household CRRA, owns all firms

$$C = \left(\int_0^1 c(j)^{\frac{\mu-1}{\mu}} dj \right)^{\frac{\mu}{\mu-1}}$$

Industry j one good, Cournot

$$y(j) = \sum_{i=1}^{n} y_i(j)$$

Technology labour only

$$y_i(j) = A_i(j)\, l_i^p(j)$$

Labour supplied inelastically

$$1 = L^p + L^r$$

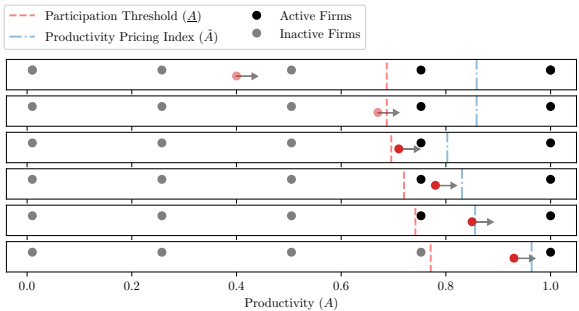
Who produces is set by the price, not by fixed costs

$$\text{firm } i \text{ produces} \iff A_i > \underline{A} = \frac{\tilde{A}}{m}, \qquad m = \frac{\mu\tilde{n}}{\mu\tilde{n} - 1}$$

$\tilde{A}$: harmonic mean of the active firms' productivities; $\tilde{n}$: number of active firms; m: price over their mean marginal cost.

- A unique equilibrium: the $\tilde{n}$ most productive firms produce. No fixed costs or entry rules are needed.
- μ is the elasticity of demand: in every industry, Lerner index = HHI/μ.
- Few firms produce: participation grows only like $\sqrt{2n/\mu}$.

Entry and exit as corner solutions



One firm (red) improves while five rivals stay fixed. It enters at $\underline{A}$ as the least productive producer; as it improves, $\underline{A}$ rises until a rival exits: $2 \to 3 \to 2$ producers.

Kuhn–Tucker condition

$$\left(1 - \frac{s_i}{\mu}\right) p\, A_i - w \;\leq\; 0$$

with equality if $y_i > 0$; $s_i = y_i / y$

Best response

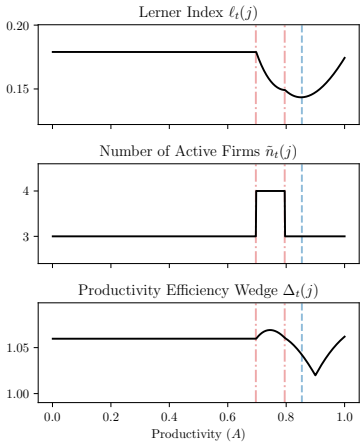
$$y_i = \max\left\{0,\; \mu\, y\left(1 - \frac{w}{p\, A_i}\right)\right\}$$

The corner

$$y_i > 0 \iff A_i > \underline{A} = w/p$$

the price covers the firm's marginal cost

Entry lowers markups but wastes productivity



The price over the efficient price has two wedges:

$$\zeta(j) = \underbrace{\mathcal{M}(j)}_{\text{markup}} \times \underbrace{\Delta(j)}_{\text{efficiency wedge}}$$

$\Delta(j) = A_{\max}(j)/\bar{A}(j) \geq 1$: frontier over realised productivity.

- At entry (first red line) the Lerner index starts to fall …
- … and Δ starts to rise: the entrant is the least productive producer.

Innovation builds on the technology in use

$$\eta = 1 - \exp\big(-\bar{\eta}\,(l^r)^{\theta}\,a^{-\varepsilon}\big), \qquad a = \frac{A_i}{\bar{A}}, \qquad A_i' = \gamma A_i \text{ on success}$$

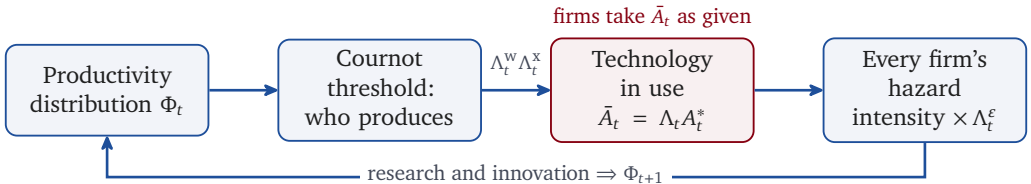
θ	research elasticity: how strongly success responds to research labour
ε	catch-up: firms behind the baseline innovate more easily, firms ahead less
$\bar{\eta}$	scale of innovation arrivals
γ	size of a step

The key assumption. The baseline is technology *in use*, $\bar{A} = Y/L^p$: output per production worker. It depends on who produces, and each firm takes it as given: an **externality**.

The dynamic inefficiency of competition

$$Y_t = A_t^* \; \Lambda_t \; (1 - L_t^r), \qquad \bar{A}_t = \Lambda_t \, A_t^*, \qquad \Lambda_t \leq 1$$

A_t^*: the frontier technology. Λ_t: static efficiency, lost to sub-frontier producers and to markup dispersion.



Misallocation today lowers every firm's chance to innovate tomorrow.

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Two regimes, one model



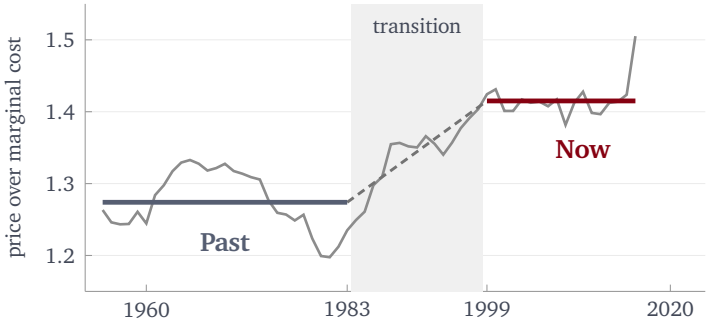
Fixed in both regimes

$n = 3$ potential producers per industry, $\beta = 0.97$, $\sigma = 2$. Each regime is a balanced growth path.

Targets: the regime means of

the Lerner index, business-funded R&D / GDP, and TFP growth.

The regimes are dated from the aggregate markup

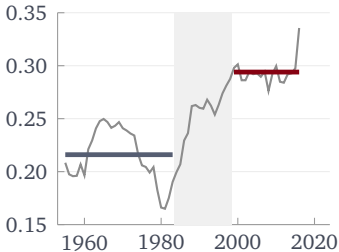


- Fit a ramp (flat, linear, flat); least squares picks the break years.
- Breaks in **1983** and **1999**; near-optimal 1980–85 and 1995–2004.
- Unchanged when 2015–16 are dropped.

The transition opens with deregulation, the Volcker disinflation and Bayh–Dole (1980), and closes with the ICT boom and China's entry to the WTO (2001). Markup: total-cost-weighted, U.S. listed firms (De Loecker et al., 2020); dashed: the fitted ramp.

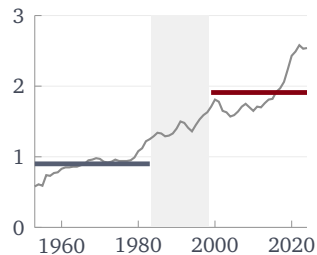
Three targets: markups up, R&D doubled, growth down

Lerner index



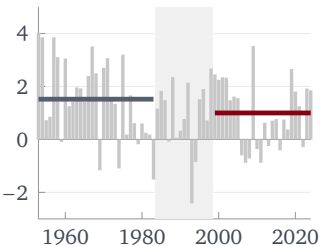
$0.216 \rightarrow \mathbf{0.294}$

Business R&D / GDP (%)



$0.90 \rightarrow \mathbf{1.91}$

TFP growth (% a year)



$1.52 \rightarrow \mathbf{1.00}$

Thick lines: regime means. Shaded: the transition, not targeted. Annual TFP growth is volatile, so its fall is imprecisely measured ($p = 0.23$).

Data: De Loecker et al. (2020); NSF NCSES (2026); Fernald (2014).

The model matches every target in both regimes

	Past			Now		
Target	**Data**	**Model**	z	**Data**	**Model**	z
Lerner index	0.216	0.213	−0.34	0.294	0.295	+0.40
	(0.009)			(0.003)		
Business R&D / GDP (%)	0.90	0.91	+0.17	1.91	1.88	−0.34
	(0.05)			(0.11)		
TFP growth (%)	1.52	1.51	−0.03	1.00	1.16	+0.58
	(0.29)			(0.28)		

Data: regime means, with Newey–West standard errors in parentheses. z = (model − data)/s.e. Model: simulated $n = 3$ economies (runs #2001 and #2010).

All six moments are within 0.6 standard errors.

Beyond the targets: right direction, too small

Not targeted	Data	Model
Markup dispersion[a]	0.9% → 4.2%	0.5% → 1.2%
Producers per industry (of 3)	—	3.0 → 2.5
Establishment entry rate	13.4% → 10.6%	not comparable[b]
Within-industry s.d. of log TFP	0.38 → 0.47	at most 0.17[c]

[a] Sales-weighted over cost-weighted markup, minus one. [b] In the model a firm that stops producing can restart; the BDS counts establishment births. [c] Anywhere in the $n = 3$ parameter sweep.

- Markup dispersion rises 2.5-fold in the model, almost 5-fold in the data.
- Entry, exit and dispersion need more heterogeneity than three producers allow.

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The structural change between the regimes

	Parameter	Past	Now	Change	Order	Meaning
θ	research elasticity	0.050	0.132	**×2.6**	**first**	**More research, less per unit**
ε	catch-up	2.01	1.17	**−42%**	**second**	**Less catch-up, longer leads**
μ	demand elasticity	1.70	1.51	−12%	—	Higher markup at given HHI
$\gamma - 1$	step size	3.8%	3.2%	−15%	—	Smaller step per success
$\bar{\eta}$	hazard scale	0.65	0.81	+24%	—	Level shift, offsets θ

Rows are ordered by how much of the shift each parameter can carry on its own ▸ **backup**. ε, γ and $\bar{\eta}$ are only weakly identified by three targets.

First order: θ rose. Second order: ε fell.

Research matters more, relative position matters less

$$\eta_i = 1 - \exp\Big(- \bar{\eta} \underbrace{\left(l_i^r\right)^{\theta}}_{\text{research effort}} \underbrace{\left(\frac{A_i}{\bar{A}}\right)^{-\varepsilon}}_{\text{relative position}} \Big)$$

θ 0.050 → 0.132 — **first order**

- Success responds more strongly to research effort.
- So extra research pays more: R&D / GDP doubles.
- But at firms' small research levels, the same effort buys less ▸ **backup**.

ε 2.01 → 1.17 — **second order**

- Success depends less on a firm's position relative to $\bar{A}$.
- Laggards catch up less easily; leaders face less fishing-out.
- So leaders keep their lead: fewer producers (3.0 → 2.5), higher markups.

Takeaways

1. **Entry and exit as corner solutions.** Who produces, and how that changes, is an equilibrium outcome, without fixed costs or entry rules.

2. **Innovation relative to the technology in use.** Entry lowers markups but pulls production off the frontier, which slows every firm's research: a dynamic cost of competition.

3. **The U.S. application.** One model fits the Past and the Now. First order, θ almost tripled: firms research more but get less per unit. Second order, ε fell: leaders keep their lead, and fewer firms produce.

Next steps

- **Decompose the change with solved counterfactuals**: the Past economy with one Now parameter at a time.
- **What the model misses**: the rise in markup dispersion, and entry and exit (larger n).
- **Policy under the Now technology**: R&D subsidies, competition policy, and the planner's benchmark.

Thank you

Questions and comments welcome

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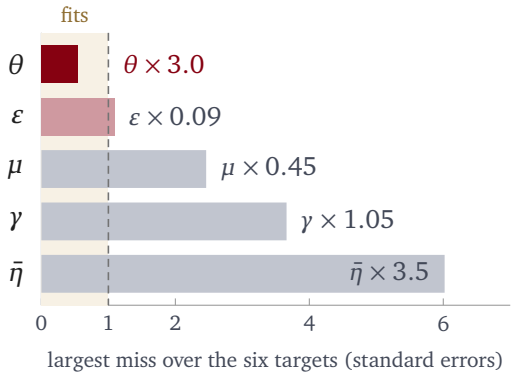
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Backup: θ comes first, ε second

back



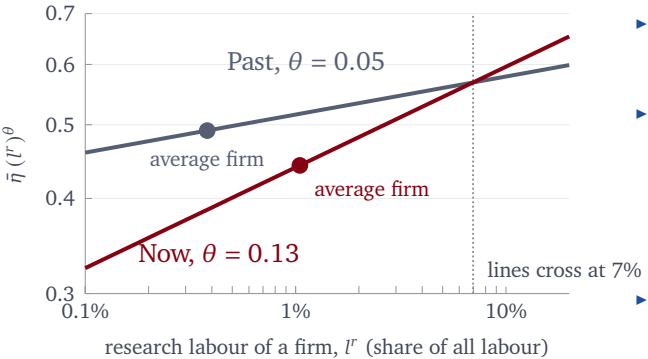
Let one parameter differ between the regimes, keep the other four common, and refit.

- Only θ keeps all six targets within one standard error. It needs $\theta \times 3.0$; the calibration has $\times 2.6$.
- ε comes next: alone it just misses. Without θ it would have to fall by 81–92%; the calibration has −42%.

Emulators of the $n = 3$ model, fitted to a 1,000-draw parameter sweep; subsets of up to three parameters.

Backup: a higher θ, more research but less per unit

back



- **Steeper**: extra research pays more, so firms do more of it.
- **Lower** where firms operate, well below 7%: the same effort buys less. The average Now firm does 2.8 times the research and still scores lower.
- In the model: growth 1.51% → 1.16%, producers 3.0 → 2.5, markups up.

Firm at the baseline ($a = 1$); log scales, slope = θ. Dots: research per active firm, $L^r/\tilde{n}$. Compare Bloom et al. (2020).

Backup: data sources

Series	Source	Years	Use
TFP growth, utilization-adjusted	Fernald (2014)	1948–2025	target
Business-funded R&D / GDP	NSF NCSES (2026)	1953–2024	target
Aggregate markup, total-cost weights	De Loecker et al. (2020)	1955–2016	target
Net operating surplus / GDI	BEA (2026)	1959–2025	check
Establishment entry and exit	Census BDS (2025)	1978–2023	not targeted
Within-industry dispersion of log TFP	DiSP (2026)	1987–2021	not targeted

The rebuilt markups reproduce DEU's 1.21 (1980) and 1.61 (2016) and the indices of Albrecht and Decker (2024). The R&D tables pass twelve accounting checks from the NSF report.

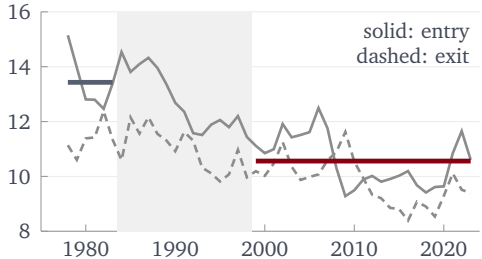
Backup: the regime dates

Series	T_1	near-optimal	T_2	near-optimal
Markups jointly (baseline)	1983	1980–85	1999	1995–2004
Markups jointly, without 2015–16	1983	1980–85	1999	1994–2003
Markup, total-cost-weighted	1984	1981–86	1999	1995–2005
Markup, sales-weighted	1983	1980–86	2003	1998–2012
Business-funded R&D / GDP	1967	1958–75	2020	2019–20
Establishment entry rate	1985	1983–89	2011	2007–17

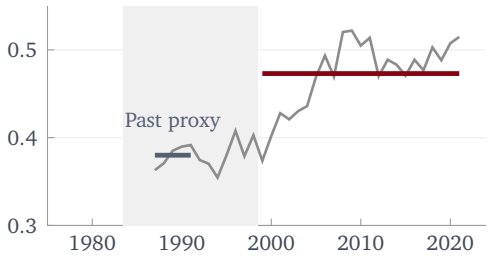
Least-squares dates of the ramp: flat to T_1, linear to T_2, flat after. The markup series are standardised and share the dates. Near-optimal: within a factor $1 + 5.99/N$ of the minimum sum of squared residuals. R&D rises gradually over the whole sample, and entry declines later, so neither dates the regimes.

Backup: dynamism down, dispersion up

Entry and exit (% a year)



Within-industry s.d. of log TFP

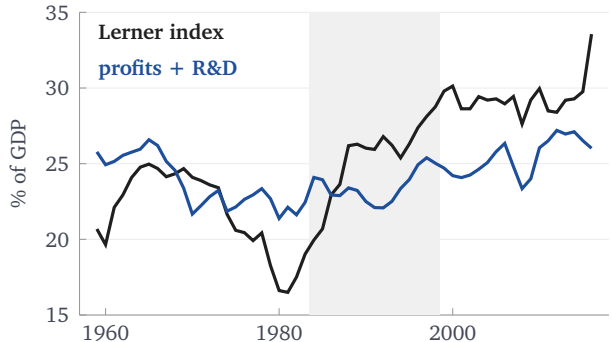


- Entry 13.4% → 10.6%, exit 11.4% → 9.8%: expected age at exit 8.8 → 10.3 years.
- Dispersion is flat to the late 1990s, then rises, most in the upper tail.

Census BDS (2025); DiSP (2026), 86 manufacturing industries.

Backup: rents equal profits plus R&D, on average

$$\mathscr{L} = \pi + \frac{wL^r}{Y} : \quad \text{markup rents are paid out or spent on R\&D}$$



- 1959–2016: profits and R&D are 97% of rents; gap +0.9 pp (s.e. 0.8).
- Not year by year: −1.6 pp before 1990, +3.7 pp after.
- The gap tracks overhead (SG&A), absent from the model: $R^2 = 0.81$.

Profits: net operating surplus over GDI (U.S. Bureau of Economic Analysis, 2026); R&D: business-funded.

Backup: which markup, and the markup gap

- The model's aggregate markup is total sales over total cost: a **cost-weighted** mean of firm markups. The target uses DEU markups with total-cost weights (COGS, overhead and capital).
- The sales-weighted markup adds one moment, dispersion:

$$\frac{M^{\text{sales}}}{\mathcal{M}} - 1 = \frac{\text{Var}_{\text{cost}}(m_i)}{\mathcal{M}^2} \geq 0.$$

- In the data this gap rises from 0.9% to 4.2%. Two-thirds of the 1980–2016 rise in the sales-weighted markup is reallocation towards high-markup firms.

Backup: finding which parameters must move

1. **Sweep**: 1,000 draws of $(\mu, \gamma, \theta, \varepsilon, \bar{\eta})$, solved at $n = 1$ and 2 (848 and 752 usable) and, so far, 74 at $n = 3$.
2. **Emulators**: Gaussian processes for log growth, log R&D / GDP and logit Lerner. At $n = 3$: the $n = 2$ emulator, extrapolated by the $n = 1 \to 2$ step, plus a correction fitted to the $n = 3$ draws.
3. **Implausibility**: $I = |\text{data} - \text{emulator}| \Big/ \sqrt{\text{s.e.}^2 + \text{s.d.}^2_{\text{emulator}}}$.
4. **Subsets**: let only some parameters differ between Past and Now. A subset explains the shift if all six targets have $I \leq 1$ in competitive industries.

The calibrated economies themselves (runs #2001 and #2010) are solved directly, not emulated.

Backup: the equilibrium active set

A set $\mathcal{N}^*$ of $\tilde{n}$ firms is an equilibrium active set iff it is internally and externally consistent:

$$A_i > \frac{\mu\tilde{n} - 1}{\mu\tilde{n}}\,\tilde{A}(\mathcal{N}^*) \quad \forall i \in \mathcal{N}^*, \qquad A_{i'} \le \frac{\mu(\tilde{n}+1) - 1}{\mu(\tilde{n}+1)}\,\tilde{A}(\mathcal{N}^* \cup \{i'\}) \quad \forall i' \notin \mathcal{N}^*.$$

Corollary (constructive characterisation). Order $A_{(1)} \ge \cdots \ge A_{(n)}$ and let $F(k) = A_{(k)} \sum_{i \le k} 1/A_{(i)} - k + 1/\mu$. Then $F(1) = 1/\mu > 0$, F is non-increasing, and $\tilde{n} = \max\{k : F(k) \ge 0\}$.

- Monopoly with no fixed cost iff $A_{(2)} \le \frac{\mu-1}{\mu} A_{(1)}$.
- The search over 2^n subsets reduces to a sort and one $O(n)$ scan.

Backup: an exact decomposition of output

Proposition (static efficiency decomposition). At every productivity distribution,

$$Y_t = \underbrace{A_t^*}_{\text{technology available}} \times \underbrace{\Lambda_t^{\mathrm{w}}}_{\text{sub-frontier producers}} \times \underbrace{\Lambda_t^{\mathrm{x}}}_{\text{markup dispersion}} \times \underbrace{(1 - L_t^r)}_{\text{labour in production}}$$

$$\Lambda^{\mathrm{w}} = \left(\int_0^1 \omega^*(j)\, \Delta(j)^{-(\mu-1)} dj \right)^{\frac{1}{\mu-1}} \leq 1$$

Equal to one iff only frontier firms produce.

$$\Lambda^{\mathrm{x}} = \frac{\left(\int_0^1 \omega(j)\, \mathcal{M}(j)^{-(\mu-1)} dj \right)^{\frac{\mu}{\mu-1}}}{\int_0^1 \omega(j)\, \mathcal{M}(j)^{-\mu} dj} \leq 1$$

Equal to one iff markups are equal across industries.

A_t^*: the frontier index. The markup *level* does not appear; only its dispersion costs output.

Backup: the stationary problem

Normalised law of motion, with $1 + g_{t+1} = \bar{A}_{t+1}/\bar{A}_t$:

$$a_{i,t+1} = \begin{cases} \dfrac{\gamma}{1+g_{t+1}}\, a_{i,t} & \text{w.p. } \eta(l^r_{i,t}, a_{i,t}), \\ \dfrac{1}{1+g_{t+1}}\, a_{i,t} & \text{otherwise.} \end{cases}$$

Stationary Bellman equation:

$$\hat{V}(\mathbf{a}_i) = \max_{l^r} \left\{ \hat{d}_i(\mathbf{a}_i, l^r) + \beta(1+g)^{1-\sigma}\, \mathbb{E}\, \hat{V}(\mathbf{a}_i') \right\}$$

- Symmetric Markov-perfect equilibrium; each industry is an Ericson and Pakes (1995) game, linked to the others only through the aggregates.
- g and the invariant distribution are a joint fixed point.
- Moments: 5,000 industries simulated for 4,000 periods, after a burn-in of 3,000.

References I

Aghion, P., Bergeaud, A., Boppart, T., Klenow, P. J., and Li, H. (2023). A theory of falling growth and rising rents. *Review of Economic Studies*, 90(6):2675–2702.

Aghion, P., Bloom, N., Blundell, R., Griffith, R., and Howitt, P. (2005). Competition and innovation: An inverted-U relationship. *Quarterly Journal of Economics*, 120(2):701–728.

Aghion, P., Harris, C., Howitt, P., and Vickers, J. (2001). Competition, imitation and growth with step-by-step innovation. *Review of Economic Studies*, 68(3):467–492.

Aghion, P. and Howitt, P. (1992). A model of growth through creative destruction. *Econometrica*, 60(2):323–351.

Akcigit, U. and Ates, S. T. (2023). What happened to U.S. business dynamism? *Journal of Political Economy*, 131(8):2059–2124.

Albrecht, B. C. and Decker, R. A. (2024). Rising markups and declining business dynamism: Evidence from the industry cross section. FEDS Notes, Board of Governors of the Federal Reserve System.

Autor, D., Dorn, D., Katz, L. F., Patterson, C., and Van Reenen, J. (2020). The fall of the labor share and the rise of superstar firms. *Quarterly Journal of Economics*, 135(2):645–709.

References II

Benkard, C. L., Miller, N., and Yurukoglu, A. (2026). The rise of market power and the macroeconomic implications: Comment. Technical report, Graduate School of Business, Stanford University and NBER.

Bloom, N., Jones, C. I., Van Reenen, J., and Webb, M. (2020). Are ideas getting harder to find? *American Economic Review*, 110(4):1104–1144.

Cavenaile, L., Celik, M. A., and Tian, X. (2026). Are markups too high? competition, strategic innovation, and industry dynamics. *Review of Economic Studies*. Forthcoming.

De Loecker, J., Eeckhout, J., and Unger, G. (2020). The rise of market power and the macroeconomic implications. *Quarterly Journal of Economics*, 135(2):561–644.

De Ridder, M. (2024). Market power and innovation in the intangible economy. *American Economic Review*, 114(1):199–251.

Ericson, R. and Pakes, A. (1995). Markov-perfect industry dynamics: A framework for empirical work. *Review of Economic Studies*, 62(1):53–82.

Fernald, J. (2014). A quarterly, utilization-adjusted series on total factor productivity. Working Paper 2012-19, Federal Reserve Bank of San Francisco.

References III

Grossman, G. M. and Helpman, E. (1991). Quality ladders in the theory of growth. *Review of Economic Studies*, 58(1):43–61.

Hopenhayn, H. and Okumura, K. (2026). Dynamic oligopoly and innovation. Working Paper; Date accessed: 2026-09-10.

Klette, T. J. and Kortum, S. (2004). Innovating firms and aggregate innovation. *Journal of Political Economy*, 112(5):986–1018.

Liu, E., Mian, A., and Sufi, A. (2022). Low interest rates, market power, and productivity growth. *Econometrica*, 90(1):193–221.

National Center for Science and Engineering Statistics (2026). National patterns of R&D resources: 2023–24 data update. Report NSF 26-313, U.S. National Science Foundation.

Peters, M. (2020). Heterogeneous markups, growth, and endogenous misallocation. *Econometrica*, 88(5):2037–2073.

U.S. Bureau of Economic Analysis (2026). Shares of gross domestic income: Net operating surplus. Retrieved from FRED, Federal Reserve Bank of St. Louis.

References IV

U.S. Bureau of Labor Statistics and U.S. Census Bureau (2026). Dispersion statistics on productivity (DiSP). Experimental data product.

U.S. Census Bureau (2025). Business dynamics statistics, 1978–2023. Released 25 September 2025.